CLASS 10 — MATHS · FORMULA SHEET

Chapter 3: **Pair of Linear Equations in Two Variables** | Quick revision, one page

Standard form and solutions

- $a_1x + b_1y + c_1 = 0, \quad a_2x + b_2y + c_2 = 0$ with $a_i^2 + b_i^2 \neq 0$.
- A solution is a pair (x, y) satisfying both equations; geometrically, a common point of the two lines.

Nature of solutions (graphical view)

- $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$: lines intersect at one point; unique solution; **consistent**.
- $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$: lines coincide; infinitely many solutions; consistent (dependent).
- $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$: lines are parallel; no solution; **inconsistent**.

Graphical method

- Draw both lines using a table of 2 or 3 points each; read the common point of intersection.
- Check by substituting the point in both equations.

Substitution method

- Express one variable in terms of the other from one equation, substitute in the other, solve, then back-substitute.

Elimination method

- Multiply the equations by suitable numbers to make the coefficients of one variable equal in magnitude; add or subtract to eliminate it.
- Solve for the remaining variable, then substitute back.
- A false statement such as $0 = 5$ means no solution; a true statement such as $0 = 0$ means infinitely many solutions.

Word problems

- Define two variables, translate two conditions into two equations, solve, and check against the original statement.
- Two-digit number: $10x + y$; reversed: $10y + x$.

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